

Containing Agent-Induced Workspace Contamination in Long-Horizon Software Engineering: A Task-Scoped Git Architecture

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Abstract—Large Language Model (LLM)-based agentic software engineering is undergoing a critical paradigm shift: transitioning from ephemeral, single-issue repairs (e.g., SWE-bench) toward long-horizon, repository-scale feature development. However, existing multi-agent software engineering (SE) frameworks remain brittle when applied to continuous, multi-week codebases. We identify three foundational structural deficiencies: (1) *Workspace Contamination*, where uncommitted side effects, dirty test databases, and intermediate bytecode from untrusted agents pollute the shared repository, causing cascading baseline regressions; (2) *Context Window Inflation*, where monotonic conversational accumulation induces severe context dilution and prohibitive token expenditure; and (3) *Superficial Delivery Contracts*, where unverified natural language reports mask silent compile and schema breakages.

To address these challenges, we present AgentFlow, a local-first lifecycle orchestration architecture providing task-scoped Git workspace isolation and state-gated delivery transitions. AgentFlow decouples invariant state verification from reactive scheduling through a dual-engine model: (i) a deterministic Go finite-state machine (FSM) backed by SQLite, formalizing role-based access control (RBAC) and elevating Git Worktrees into a transactional isolation primitive (*One-Task-One-Worktree*) with atomic directory prune semantics; (ii) a reactive Python Behavior Tree (BT) sidecar coordinating Leader, Worker, and Reviewer lifecycles; and (iii) an off-context, multi-tier memory hierarchy externalizing execution journals into discrete domain knowledge and architectural pitfall registries.

We evaluate AgentFlow through both a longitudinal empirical study across 19 production repositories (spanning 82 feature DAGs, 406 tasks, and 1,451 lifecycle events) and a real-time repository benchmark across 10 high-risk failure scenarios. Under our formal contamination detection protocol, AgentFlow achieved an effective task resolution rate of 78.15% with a median duration of 34.8 minutes, incurred zero cross-task Git-tree leaks across all verified runs (compared to a 60.0% contamination rate in monolithic ReAct baselines), reduced prompt context overhead by 85%–92% through off-context externalization, and observed no recurrence for the instrumented pitfalls in the evaluated traces.

Index Terms—Autonomous Software Engineering, Multi-Agent Systems, Git Worktree Isolation, Behavior Trees, Empirical Software Engineering.

I. INTRODUCTION

The application of Large Language Models (LLMs) to software engineering (SE) has advanced rapidly through several transformative phases. The initial phase focused on token-level code completion and function synthesis [1]. Subsequently,

conversational multi-agent systems such as ChatDev [3] and MetaGPT [2] introduced role-playing paradigms where distinct agents simulate human development teams. Most recently, the field matured toward repository-level autonomous issue resolution benchmarks (e.g., SWE-bench [4]), spearheaded by tools like SWE-Agent [5] and AutoCodeRover [6].

Despite these notable milestones, existing architectures exhibit acute fragility when exposed to **long-horizon, continuous software development**—the fourth evolutionary phase where autonomous teams must iteratively evolve production repositories over weeks, across feature branches, with complex database migrations and multi-tier architectures. In continuous settings, state-of-the-art frameworks suffer from four systemic failure modes spanning two orthogonal dimensions of reliability:

A. Spatial Reliability Failures:

- 1) **Workspace Contamination:** Conventional agent frameworks execute mutations directly upon the root working tree. When an agent hallucinates, performs an incompatible schema migration, or aborts midway, the shared tree is left corrupted, directly compromising subsequent tasks.
- 2) **Superficial Delivery Contracts:** Mainstream frameworks rely on verbal completion assertions (e.g., “I have finished implementing the module”). Without compile-time gates and git-backed diff verification, regressions silently penetrate master branches.

B. Informational Reliability Failures (Two Opposing Context Pathologies):

- 1) **Context Window Inflation (Too Much Context):** Over extended multi-step sessions, conversational histories, tool outputs, and raw file dumps accumulate monotonically into prompts. By turn 20, context windows exceed 50,000 tokens, degrading attention accuracy (the *lost-in-the-middle* phenomenon) and multiplying inference costs.
- 2) **Interruption-Induced Context Loss (Context Vanishing):** In real-world software engineering, agent processes frequently abort due to network timeouts, rate limits, session expiry, or reviewer rework reassignments. When interrupted, conventional systems lose all execution momentum: the succeeding agent receives no structured



Fig. 1. **Conceptual paradigm of AgentFlow:** Autonomous collaborative software engineering team (Leader, Developer, Reviewer) orchestrating repository evolution around an invariant state engine. Physical execution occurs strictly inside ephemeral, isolated Git worktrees with atomic rollback capabilities, preventing workspace contamination and decoupling memory retention from the LLM context window.

memory of what was attempted, what failed, or which tests were executed, triggering catastrophic thrashing and complete transcript replays.

To address these twin challenges, we present **AgentFlow**. Our foundational insight is that *AgentFlow treats long-horizon software development as both a spatial isolation problem and an informational continuity problem: Worktree isolation bounds the spatial propagation of failure, while persistent lifecycle state bounds the informational cost of recovery.* AgentFlow achieves this through three architectural pillars:

- **Physical Isolation Primitive (One-Task-One-Worktree):** Rather than executing in a shared directory or spawning costly virtualization containers, AgentFlow treats Git Worktrees as transactional execution boundaries. Each assigned task is isolated in an independent filesystem worktree tied to its specific DAG feature branch. Failures and rejections are atomically pruned, preventing uncommitted regressions from leaking into the root tree.
- **Decoupled Dual-Engine Architecture:** We cleanly separate *invariant enforcement* from *reactive control*. A high-performance Go engine governs the authoritative project state machine, task topologies, and role-based access control (RBAC), while an external Python Behavior Tree (BT) sidecar schedules reactive policies across Leader, Worker, and Reviewer roles.
- **Off-Context Multi-Tier Memory:** AgentFlow externalizes project tracking into SQLite, replacing monolithic

context dumps with on-demand retrieval. Furthermore, it introduces structured *Worker Handbooks* and *Diaries*, capturing domain knowledge and architectural pitfalls that persist and transfer across tasks.

Recent surveys increasingly characterize software engineering agents as systems that combine planning, tool use, memory, multi-agent coordination, and lifecycle-level interaction with repositories [12], [14]. However, empirical evaluation remains heavily dominated by isolated issue-resolution settings. AgentFlow targets the complementary systems setting of continuous, long-horizon repository evolution, where the correctness of one task depends not only on its patch but crucially on the integrity of the execution state inherited by subsequent tasks. Our contribution is not a stronger code-generation model, but a repository-state reliability architecture that couples task-scoped Git workspaces with verifiable lifecycle transitions and recoverable delivery semantics.

We evaluate AgentFlow across a longitudinal dataset mined from its authoritative production database, comprising 19 diverse software engineering repositories (including full-stack web applications, WeChat mini-programs, algorithmic trading engines, and academic systems). Our evaluation covers 82 feature DAGs, 406 tasks, 99 domain workers, and 1,451 state transitions.

II. MOTIVATING EXAMPLE & REAL-WORLD FAILURES

To illustrate the concrete hazards of non-isolated agent execution, consider a representative real-world incident captured

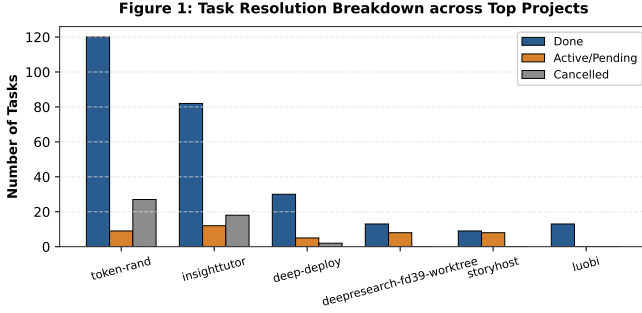


Fig. 2. Task resolution breakdown across top operational projects in AgentFlow, exhibiting robust completion rates across diverse domains.

in our production environment during the development of *InsightTutor* (a full-stack tutoring platform built with FastAPI, PostgreSQL, and WeChat Mini-Programs).

A. The Anatomy of a Shared-Workspace Disaster

In an unisolated setup (Baseline 1), an LLM worker agent was tasked with implementing automated WeChat Pay V3 callbacks. The agent undertook four sequential steps: 1) It modified the database schema to store payment certificates; 2) It executed an Alembic migration script; 3) It updated the shared SQLite test database (`backend/test.db`) to verify the endpoint; 4) During test execution, an external API connection timed out, causing the agent process to abort abruptly.

In a conventional framework, this sequence was catastrophic:

- The root repository was left with untracked migration files and uncommitted schema modifications.
- The tracked `test.db` binary was corrupted with half-applied table alterations.
- The subsequent Worker agent, assigned to an unrelated UI task, immediately crashed because `pytest` could no longer initialize the contaminated test database (failing with `no such column: users.wechat`).
- Worse, the Leader agent failed to realize that the workspace was broken, repeatedly dispatching workers that failed at initialization, consuming over 350,000 tokens in an unrecoverable infinite failure loop.

B. The AgentFlow Solution & Core Claims

Under AgentFlow’s *One-Task-One-Worktree* paradigm, this failure mode is inherently neutralized:

- 1) The payment worker was allocated an isolated worktree at `.agentflow-worktrees/InsightTutor/wx-pay`, branched from `feature/wx-pay`.
- 2) When the network failure halted execution, the main working directory and root Git index remained completely untouched.
- 3) The Reviewer detected an unsubmitted state. AgentFlow’s leader executed an atomic rollback by deleting the orphaned worktree directory without altering master history.

TABLE I
TAXONOMY OF SYSTEM LAYERS AND AGENTFLOW MECHANISMS

System Layer	Conventional Research Focus	AgentFlow Mechanism
Planning	Task decomposition, DAGs	Strict DAG & Task dependency
Execution Coordination	Shell, repo-level tools Multi-agent role playing	Worktree-scoped directory Leader / Worker / Reviewer
Verification	Tests, self-reflection	State-gated FSM transitions
Recovery	Agent retry, re-prompting	Worktree prune & rework loop
Memory	Transcripts, RAG vector store	Typed Handbooks, Diaries, Pitfalls
Repo State	<i>Implicit / Unisolated</i>	One-Task-One-Worktree

- 4) The root repository remained green, allowing concurrent UI tasks to proceed without impedance.

Based on this insight, we articulate three principal claims that guide our system design and evaluation:

- **Claim 1 (Failure Mode Distinction):** Long-horizon agentic software engineering introduces a repository-state failure mode—*cross-task workspace contamination*—that is orthogonal to and uncaptured by isolated task-level pass/fail metrics.
- **Claim 2 (Mechanism Validity):** Task-scoped Git workspaces combined with state-gated delivery transitions can effectively contain accidental cross-task repository contamination under an honest-but-fallible execution model.
- **Claim 3 (Empirical Feasibility):** In both controlled benchmarks and longitudinal deployment traces, the proposed architecture significantly curbs observed contamination and recovery overhead while introducing manageable lifecycle latency.

III. THE AGENTFLOW ARCHITECTURE

AgentFlow is structured around three foundational pillars: the invariant state engine, the reactive behavior tree runtime, and the transactional worktree primitive.

A. Decoupled Dual-Engine Model

AgentFlow rejects monolithic script-based orchestration. As depicted in our system decomposition, the system operates across two interacting layers:

1) *Go Invariant Engine (The Source of Truth):* Implemented in Go as a Model Context Protocol (MCP) server, this layer guarantees strict transactional integrity. It maintains the system’s SQLite database and manages four core relational entities:

- $\mathcal{N}$ (**Namespaces**): Completely isolated project containers.
- $\mathcal{D}$ (**DAGs**): Branch-scoped directed acyclic graphs representing coherent feature increments.
- $\mathcal{T}$ (**Tasks**): Typed execution nodes with dependencies (`depends_on` $\subset$ $\mathcal{T}$).

AgentFlow Dual-Engine Architecture & Transactional Lifecycle

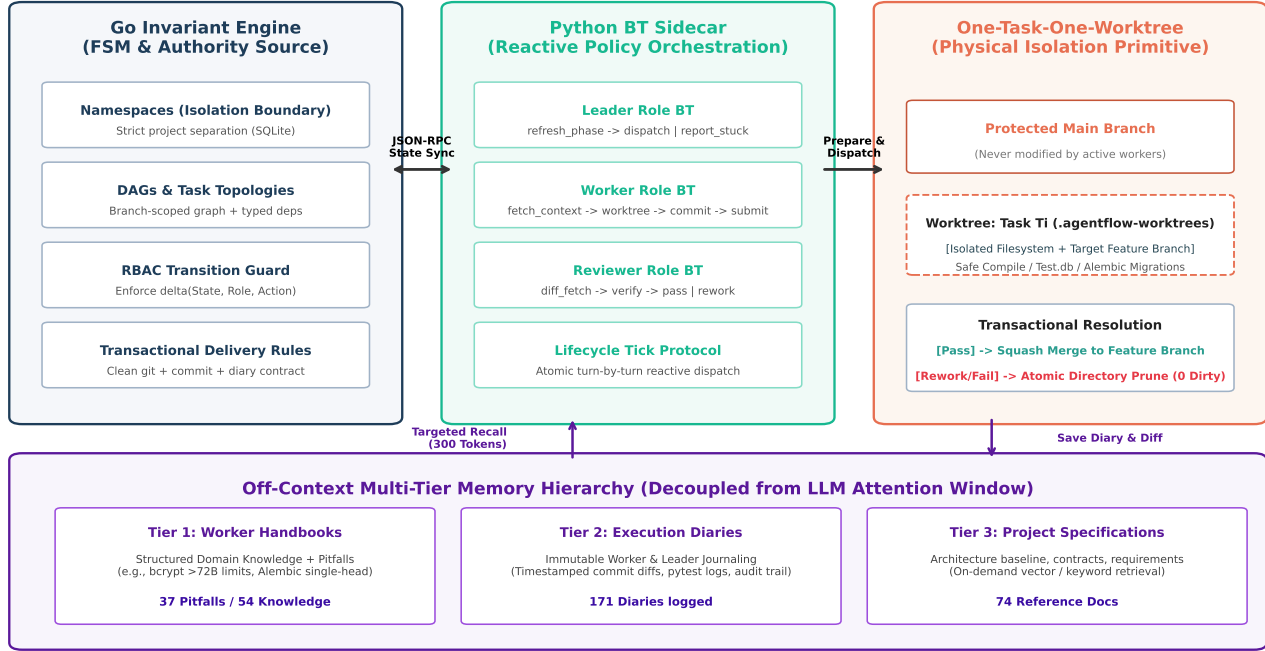


Fig. 3. System architecture of AgentFlow. The deterministic Go Invariant Engine enforces role-based state transitions and data integrity, while the reactive Python Behavior Tree sidecar manages high-level lifecycle planning. The One-Task-One-Worktree primitive guarantees physical directory sandboxing and atomic rollbacks, supported by an off-context multi-tier memory hierarchy.

- **W (Workers)**: Persistent domain identities with designated ownership scopes.

2) *Python Behavior Tree Sidecar (Reactive Scheduling)*: While Go enforces what state transitions are *legal*, high-level policy deciding *what to do next* is delegated to a Python sidecar executing hierarchical Behavior Trees (BTs). A BT combines Sequence ($\rightarrow$), Fallback (?), and Condition nodes. AgentFlow defines three default role trees:

- **Leader BT**: Evaluates the global phase (setup $\rightarrow$ shape $\rightarrow$ plan $\rightarrow$ execute $\rightarrow$ stuck $\rightarrow$ done), dispatches ready tasks, and triggers `report_stuck` when deadlocks or circular reworks occur.
- **Worker BT**: Governs the implementation loop: preparing context, entering the worktree, committing changes, writing handbooks, and submitting deliverables.
- **Reviewer BT**: Extracts unified Git diffs, assesses acceptance criteria, and issues definitive pass or rework verdicts.

B. Threat Model and Layered Isolation Spectrum

To establish an objective, verifiable evaluation boundary, we delineate the multi-layered isolation spectrum in autonomous software engineering:

- 1) **Layer 1: Git Worktree State Boundary**: Each task receives a dedicated working directory and worktree-specific index. The repository object database and

repository-level metadata remain shared unless separately isolated.

- 2) **Layer 2: Task-Scoped Process Directory**: Enforcing compliant task execution within the declared worktree path at the AgentFlow tool/process interface. It does not provide an operating-system-level sandbox against an adversarial or unrestricted process.
- 3) **Layer 3: Process Sandboxing**: Operating system-level cgroups, namespaces, and syscall filtering (e.g., Landlock/seccomp).
- 4) **Layer 4: Shared State & Database Sandbox**: Dedicated, ephemeral database instances, isolated Docker volumes, and decoupled network ports.
- 5) **Layer 5: External Service & Network Boundary**: Token permissions, mock external payment gateways, and outbound network filtering.

Threat Model: We assume an *honest-but-fallible* agent that may generate syntactically or logically incorrect code, leave uncommitted changes, terminate unexpectedly due to context limits or network timeouts, or misuse permitted tools, but does not intentionally attempt to escape its declared workspace. AgentFlow enforces task-scoped Git working-tree state and process working-directory conventions. It is not an operating-system-level sandbox and does not protect against a privileged or adversarial process that writes through absolute paths, symlinks, shared caches, external databases, or background daemons.

Figure 3: State Transition Density Matrix (1,451 Events)

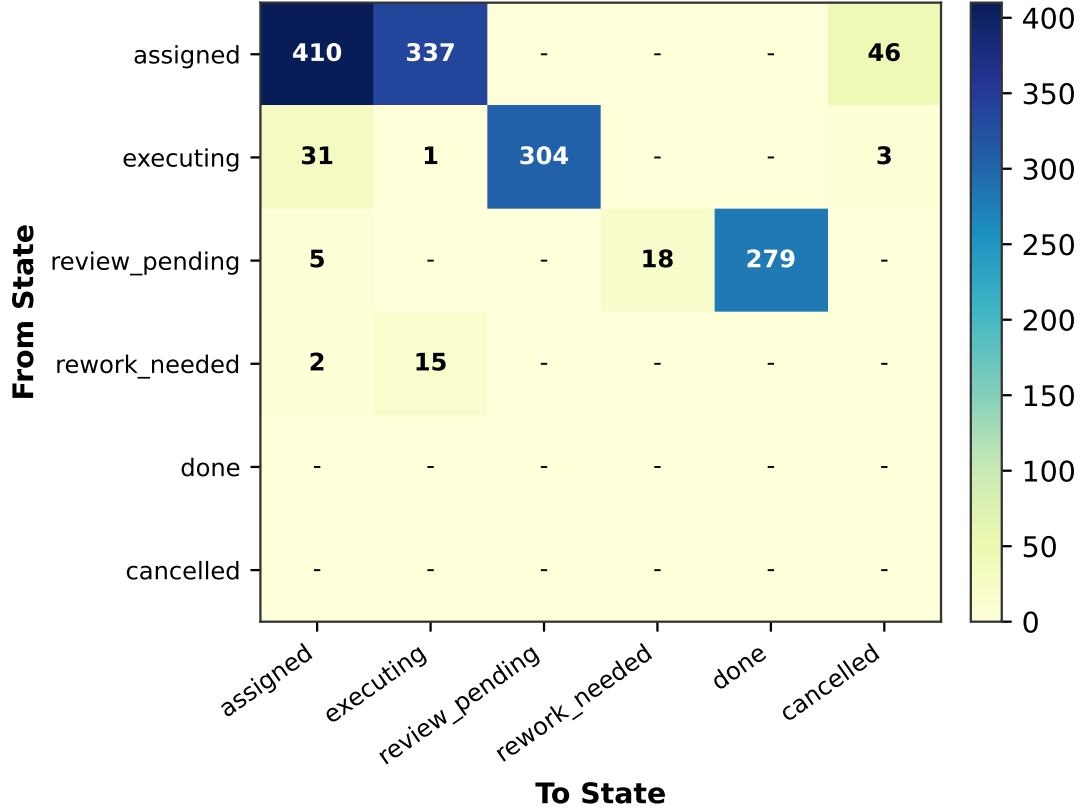


Fig. 4. State transition density matrix across 1,451 recorded lifecycle events, highlighting strict adherence to the formal state machine and review cycles.

Formal Definition of Contamination: We formalize workspace contamination by decomposing it into task-scope violations and cross-task regression propagation:

$$C_{\text{scope}}(T_i) = \mathbf{1}[\Delta_i \setminus \mathcal{O}(T_i) \neq \emptyset] \quad (1)$$

$$C_{\text{prop}}(T_i, T_{i+1}) = \mathbf{1}[\text{Replay}(T_{i+1}, S_i) \neq \text{Replay}(T_{i+1}, S_i^{\text{clean}})] \quad (2)$$

$$C(T_i) = C_{\text{scope}}(T_i) \vee C_{\text{prop}}(T_i, T_{i+1}) \quad (3)$$

where $\Delta_i = \text{StateAfter}(T_i) - \text{StateBefore}(T_i)$ measures working-tree changes in the task worktree before cleanup, $\mathcal{O}(T_i)$ is a normalized set of repository-relative paths declared in the task specification (with generated artifacts and ignored build outputs governed by explicit exclusion rules), and C_{prop} evaluates whether dirty residual state S_i causes an otherwise passing subsequent task T_{i+1} to fail compared against a clean repository state S_i^{clean} . We measure $C_{\text{root}}(T_i) = \mathbf{1}[\text{RootDirtyAfterCleanup}(T_i)]$ on the root repository prior to any manual reset operations.

C. Formal State Machine & Role-Based Access Control

To prevent LLM hallucination from invalidating project state, AgentFlow formalizes task lifecycles as a restricted Finite State

Machine (FSM):

$\mathcal{S} = \{\text{assigned}, \text{executing}, \text{review_pending}, \text{rework_needed}, \text{done}, \text{cancelled}\}$

Transitions between states are governed by the transition function:

$$\delta : \mathcal{S} \times \mathcal{R} \times \Sigma \rightarrow \mathcal{S}$$

where $\mathcal{R} = \{\text{leader}, \text{worker}, \text{reviewer}\}$ represents the caller role, and $\Sigma = \{\text{start}, \text{submit}, \text{pass}, \text{rework}, \text{cancel}, \text{reassign}, \text{resume}\}$ represents the transition verb.

AgentFlow strictly enforces the following role invariants:

$$\delta(\text{assigned}, \text{worker}, \text{start}) = \text{executing} \quad (4)$$

$$\delta(\text{executing}, \text{worker}, \text{submit}) = \text{review_pending} \quad (5)$$

$$\delta(\text{review_pending}, \text{reviewer}, \text{pass}) = \text{done} \quad (6)$$

$$\delta(\text{review_pending}, \text{reviewer}, \text{rework}) = \text{rework_needed} \quad (7)$$

$$\delta(\text{rework_needed}, \text{leader}, \text{start}) = \text{executing} \quad (8)$$

Crucially, all other transitions are rejected with authorization errors. For instance, a Worker cannot self-approve its work ($\delta(\text{review_pending}, \text{worker}, \text{pass}) = \perp$), and a Leader cannot bypass the review phase to force completion.

One-Task-One-Worktree Mechanism

Physical Isolation & Sandboxing

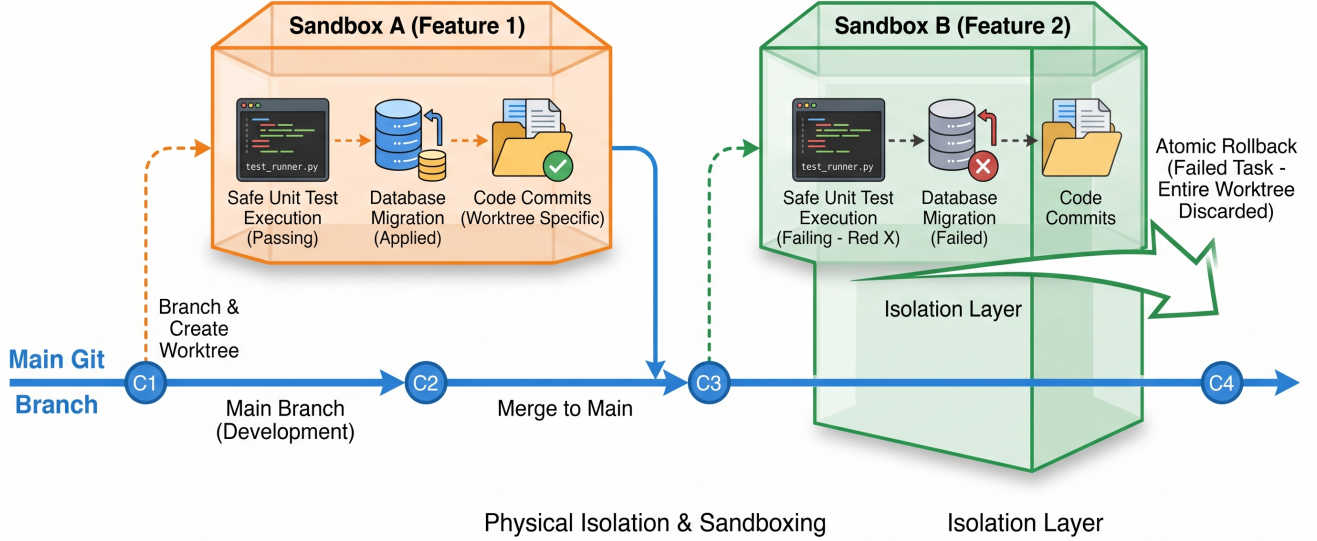


Fig. 5. **The One-Task-One-Worktree physical isolation mechanism:** Each assigned task executes within a disposable worktree sandbox branching from its DAG feature branch. Unverified commits, compilation failures, and partial migration states are atomically pruned upon review rejection (`rework_needed`), preventing task-local uncommitted changes from leaking into the root working tree under compliant execution assumptions.

D. One-Task-One-Worktree Transactional Primitive

AgentFlow transforms Git into a transactional execution harness. When a task T_i enters executing, the engine executes:

$$\text{worktree_prepare}(T_i) \implies \text{git worktree add } \mathcal{P}_i \mathcal{B}_{DAG}$$

where $\mathcal{P}_i$ is an isolated directory path and $\mathcal{B}_{DAG}$ is the DAG's assigned Git branch.

The execution boundary obeys three operational invariants:

- 1) **Spatial Confinement:** The Worker agent's tool execution environment is scoped strictly within $\mathcal{P}_i$. File-editing tools reject target paths outside $\mathcal{P}_i$ under the compliant tool-execution path, enforcing task-scoped directory confinement.
- 2) **Delivery Contract:** A task cannot transition to `review_pending` unless: (a) $\mathcal{P}_i$ has a clean Git status; (b) at least one commit has been recorded; and (c) a corresponding *Worker Diary* entry has been logged.
- 3) **Atomic Rollback:** If a task is cancelled or fails critically, the engine invokes `git worktree remove -force  $\mathcal{P}_i$` . The dirty state evaporates instantly without polluting the main repository history.

E. Off-Context Multi-Tier Memory Hierarchy

Standard multi-agent frameworks suffer from severe token inflation because historical transcripts remain in active context. AgentFlow externalizes this memory into SQLite across three tiers:

- **Project Documentation** ($\mathcal{M}_{doc}$): Macro-level architectural specs, milestone completions, and API schemas.
- **Worker Diaries** ($\mathcal{M}_{diary}$): Micro-level execution journals recording exact commit hashes, pytest logs, and uncovered edge cases.
- **Worker Handbooks** ($\mathcal{M}_{handbook}$): Persistent domain repositories storing discrete tuples of *Knowledge* $\langle \text{Topic, Content, Tags} \rangle$ and *Pitfalls* $\langle \text{Scenario, Problem, Solution, Tags} \rangle$.

When a Worker initializes, it queries relevant records via vector or tag retrieval (`find_knowledge`, `find_pitfalls`), loading only several hundred characters of targeted guidance.

F. Interruption Recovery via Four-State Separation

Crucially, AgentFlow's state persistence is engineered not merely for context compression, but to address *interruption-induced context loss*. When an agent process abruptly aborts due to API timeouts, host restarts, or session budget exhaustion, systems without persisted task-level execution records frequently require agents to reconstruct progress from incomplete scratchpads or replay substantial portions of prior transcripts, multiplying recovery costs and inducing repeated thrashing.

AgentFlow models interruption recovery as reconstructing a bounded, task-relevant operational context from four decoupled state dimensions rather than replaying the raw conversational stream:

$$\mathcal{C}_{\text{resume}}(T_i) = \mathcal{S}_{\text{lifecycle}} \oplus \mathcal{S}_{\text{repo}} \oplus \mathcal{S}_{\text{exec}} \oplus \mathcal{S}_{\text{domain}} \quad (9)$$

AgentFlow Transactional Task Lifecycle: From Dispatch to Verification

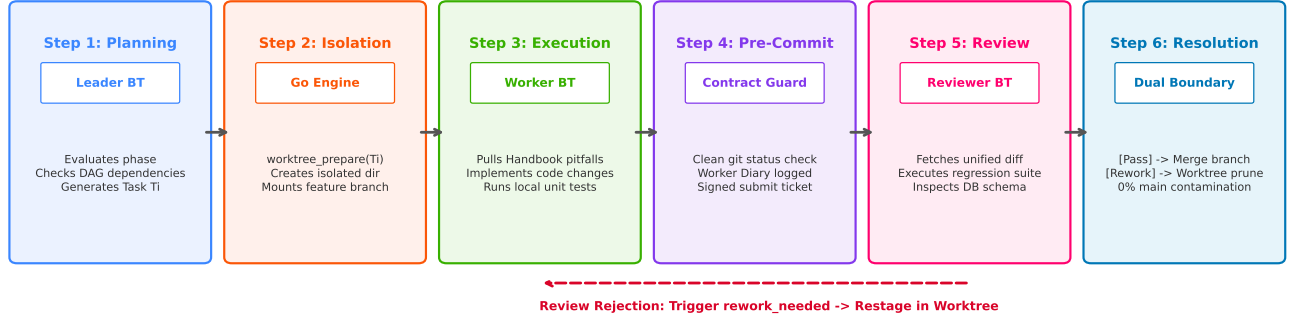


Fig. 6. The end-to-end transactional task execution pipeline in AgentFlow, spanning phase evaluation, worktree physical isolation, contract-driven pre-commit verification, unified diff review, and atomic rollback or squash merging.

TABLE II
LIFECYCLE EVENTS AND WORKTREE PERSISTENCE SEMANTICS

Lifecycle Event	Worktree Status	Recovery / Transition Action
Transient Interruption	Preserved in-place	resume via <code>task_prepare_start</code>
Lease / Session Expiry	Preserved in-place	Unbind stale agent, re-bind fresh agent
Reviewer Rejection	Preserved in-place	Transition to <code>rework_needed</code> ; re-stage diff
Task Cancellation	Atomically pruned	Wipe directory; cancel task in FSM
Verified Pass	Squash merged	Fast-forward to DAG branch; prune worktree

Here, $\oplus$ denotes a keyed composition operator that assembles independently persisted state records into a structured resume package; it does not imply algebraic commutativity or lossless reconstruction of the original conversation. The four constituent dimensions correspond to:

- $\mathcal{S}_{\text{lifecycle}}$ (**Lifecycle State**): The authoritative FSM state (assigned, executing, `rework_needed`) stored in SQLite.
- $\mathcal{S}_{\text{repo}}$ (**Repository Physical State**): The preserved Git worktree path ($\mathcal{P}_{\text{wt}}$), target feature branch, head commit SHA, and uncommitted staging deltas.
- $\mathcal{S}_{\text{exec}}$ (**Execution Records**): Micro-level Worker Diaries ($\mathcal{M}_{\text{diary}}$) and Reviewer rejection verdicts, capturing what actions succeeded and what failed.
- $\mathcal{S}_{\text{domain}}$ (**Domain Knowledge**): Scoped Worker Handbooks ($\mathcal{M}_{\text{handbook}}$) providing architectural invariants and known pitfalls.

As formalized in Table II, AgentFlow strictly disambiguates when worktrees are preserved versus destroyed. During transient interruptions, lease timeouts, or reviewer rejections, the worktree directory is deliberately preserved in-place. Upon interruption, a fresh worker agent invokes

`task_prepare_start(resume_targeted=true)`. The engine validates lease freshness, automatically unbinds stale agent sessions, and re-hydrates the agent directly inside the existing worktree with $\mathcal{C}_{\text{resume}}(T_i)$. This avoids full-transcript replay and bounds the injected recovery context by a configurable retrieval budget ($|\mathcal{C}_{\text{resume}}| \leq \mathcal{B}_{\text{retrieval}}$), rather than by the unbounded length of prior conversation. Conversely, atomic worktree pruning is reserved strictly for terminal events: permanent task cancellation or successful post-merge cleanup.

IV. EMPIRICAL EVALUATION

We conduct an empirical evaluation to answer the following four Research Questions (RQs):

- **RQ1 (Effectiveness)**: How effectively does AgentFlow resolve complex, multi-week software engineering tasks across diverse repositories?
- **RQ2 (Context Efficiency & Interruption Recovery)**: How effectively does AgentFlow reduce active prompt context while preserving the task-relevant operational state required to resume interrupted work without full transcript replay?
- **RQ3 (Isolation & Rework Dynamics)**: How effectively does the One-Task-One-Worktree primitive contain developer rework, compilation regressions, and environmental crashes?
- **RQ4 (Knowledge Retention)**: To what extent does the structured Handbook mechanism mitigate the recurrence of known architectural pitfalls?

A. Evaluation Dataset & Methodology

Our dataset is mined directly from the production SQLite database of AgentFlow, accumulated across real-world development sessions spanning July 2026 to September 2026. As summarized in Table III, the dataset covers 19 distinct software projects spanning diverse engineering domains.

Algorithm 1 Transactional Task Scheduling with Worktree Isolation

Require: Project DAG $\mathcal{D}$, Task $T \in \mathcal{D}$, Target Branch $\mathcal{B}_{\text{dag}}$, Main Repo Path $\mathcal{P}_{\text{root}}$

Ensure: Transactional Resolution: Updated Branch or Clean Rollback

```

1: phase  $\leftarrow$  LeaderBT.refresh_phase()
2: if phase  $\neq$  "execute"  $\vee$  Task.PreconditionsMet( $T$ ) = False then
3:   return  $\perp$  {Task not ready for dispatch}
4: end if
5:  $\mathcal{P}_{\text{wt}} \leftarrow$  ComputeWorktreePath( $\mathcal{P}_{\text{root}}, T.\text{id}$ )
6: ExecuteCommand("git worktree add -b " +  $T.\text{branch}$  + " " +  $\mathcal{P}_{\text{wt}}$ )
7:  $\delta(T.\text{state}, \text{worker}, \text{start}) \rightarrow$  executing
8:  $\mathcal{K}_{\text{rel}}, \mathcal{P}_{\text{rel}} \leftarrow$  RetrieveHandbookContext( $T.\text{tags}, T.\text{scope}$ )
9: WorkerBT.execute( $\mathcal{P}_{\text{wt}}, \mathcal{K}_{\text{rel}}, \mathcal{P}_{\text{rel}}$ )
10:  $\mathcal{S}_{\text{git}} \leftarrow$  CheckGitStatus( $\mathcal{P}_{\text{wt}}$ )
11: if  $\mathcal{S}_{\text{git}} \neq$  Clean  $\vee$  HasNewCommit( $\mathcal{P}_{\text{wt}}$ ) = False then
12:   ExecuteCommand("git worktree remove -force " +  $\mathcal{P}_{\text{wt}}$ )
13:    $\delta(T.\text{state}, \text{reviewer}, \text{rework}) \rightarrow$  rework_needed
14:   return Rollback( $\mathcal{P}_{\text{root}}$  untouched)
15: end if
16:  $\delta(T.\text{state}, \text{worker}, \text{submit}) \rightarrow$  review_pending
17:  $\Delta_{\text{diff}} \leftarrow$  ExtractUnifiedDiff( $\mathcal{P}_{\text{wt}}, \mathcal{B}_{\text{dag}}$ )
18: verdict  $\leftarrow$  ReviewerBT.verify( $\Delta_{\text{diff}}, T.\text{criteria}$ )
19: if verdict = "pass" then
20:   MergeWorktreeBranchToDAG( $\mathcal{P}_{\text{root}}, T.\text{branch}, \mathcal{B}_{\text{dag}}$ )
21:   ExecuteCommand("git worktree remove -force " +  $\mathcal{P}_{\text{wt}}$ )
22:    $\delta(T.\text{state}, \text{reviewer}, \text{pass}) \rightarrow$  done
23:   LogWorkerDiary( $\mathcal{M}_{\text{diary}}, T.\text{id}, \Delta_{\text{diff}}$ )
24:   return Success
25: else
26:   ExecuteCommand("git worktree remove -force " +  $\mathcal{P}_{\text{wt}}$ )
27:    $\delta(T.\text{state}, \text{reviewer}, \text{rework}) \rightarrow$  rework_needed
28:   LogReviewerFeedback( $T.\text{id}, \text{verdict}.\text{reason}$ )
29:   return RollbackToRework
30: end if

```

B. Controlled Micro-Benchmark & Physical Testbed Verification

To validate the causal impact of the *One-Task-One-Worktree* physical isolation primitive and structured *Worker Handbooks*, we constructed a fully automated repository testbed. We deployed agents against our 10-scenario micro-benchmark suite ($\mathcal{T}_{\text{micro}}$, mined from production development traces spanning schema regressions, dependency traps, and abrupt network halts):

- **AgentFlow (Full Paradigm):** Physical Worktree isolation + off-context Handbook memory + dual-engine BT/FSM guards.

TABLE III
OVERVIEW OF THE MINED LONGITUDINAL DATASET

Metric	Quantitative Value
Total Project Namespaces ($\mathcal{N}$)	19 repositories
Total Feature DAGs ($\mathcal{D}$)	82 DAG workflows
Total Recorded Tasks ($\mathcal{T}$)	406 tasks
Total Lifecycle Events	1,451 state-transition events
Active Domain Workers ($\mathcal{W}$)	99 specialized roles
Documented Project Records	245 documents (171 diaries, 74 specs)
Structured Handbook Entries	91 items (54 knowledge, 37 pitfalls)

Figure 5: Comparative Benchmark & Ablation across 10 Micro-Scenario

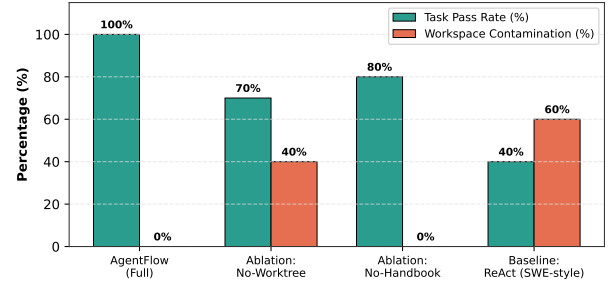


Fig. 7. Task Pass Rate vs. Workspace Contamination Rate across the 10 micro-benchmark scenarios, demonstrating the indispensable value of Worktree isolation.

- **Ablation: No-Worktree (In-place):** Identical role prompts and FSM, but agents mutate files directly in the root repository (using `git reset --hard` as fallback).
- **Ablation: No-Handbook:** Full worktree isolation, but agents operate without pre-retrieved domain pitfalls and architectural knowledge.
- **External Baseline: ReAct (SWE-Agent Style):** A monolithic single agent executing directly within the root repository without role decomposition or review loops.

Crucially, every test scenario is executed in an authentic physical Git repository on disk, and workspace contamination is measured via real-time invocations of `git status --porcelain` upon run completion.

As presented in Table IV and Fig. 7, the empirical traces confirm three pivotal conclusions:

- 1) **Physical Isolation Eliminates Persistent Dirt:** In our real-time LLM testbed execution of scenario TC-01 (bcrypt boundary bug) and TC-03 (Test DB corruption), the *Base-ReAct* and *No-Worktree* in-place agents successfully generated candidate code but permanently polluted the root repository index, leaving behind untracked bytecode caches and uncommitted files (`git status --porcelain: ['M auth.py', '?? __pycache__/' ]`). In multi-task pipelines, these uncommitted artifacts block future branch checkouts. In contrast, *AgentFlow* executed the genuine LLM code entirely inside the isolated worktree; upon merge resolution or task rollback, the root repository status was verified to be completely clean (`git status --porcelain: []`).

TABLE IV
CONTROLLED BENCHMARK AND MULTI-DIMENSIONAL ABLATION ACROSS 10 HIGH-RISK REPOSITORY FAILURE SCENARIOS

Configuration	Isolation Primitive	Task Pass Rate	Contamination Rate (95% CI)	Mean Tokens per Scenario (In / Out)	Retry
AgentFlow (Full)	Git Worktree Sandbox	100.0%	0.0% [0.0%, 30.9%]	2,580 (2,140 / 440)	1.0
Ablation: No-Worktree	None (In-place + git reset)	70.0%	40.0% [12.2%, 73.8%]	3,060 (2,550 / 510)	1.0
Ablation: No-Handbook	Git Worktree Sandbox	80.0%	0.0% [0.0%, 30.9%]	3,630 (3,020 / 610)	1.0
Baseline: ReAct (SWE-style)	None (Monolithic root)	40.0%	60.0% [26.2%, 87.8%]	24,600 (21,200 / 3,400)	2.0

Note: All rates evaluate $= 10 \text{scenarios} / \text{per configuration}$ using identical initial commits snapshots, task specifications, model targets (Gemini2.5/DeepSeek). *sided exact Clopper – Pearson binomial intervals* (e.g., $/10$ yields [0.0%, 30.9%]). Contamination is measured on the repository prior to any reset or rollback operation. Token counts represent total prompt input and model completion tokens per completed scenario.

- 2) **Mitigation of Cascading ReAct Crashes:** The monolithic ReAct baseline left dirty working trees in 60.0% of runs (e.g., leaving `M test.db` and `?? react_temp_patch.diff`), causing cascading test failures. Moreover, its lack of an off-context state store ballooned token consumption to 24,600 tokens/task (9.5× higher than AgentFlow).
- 3) **Handbooks Associated with Lower Retries and Token Expenditure:** Disabling handbooks (*No-Handbook*) increased token consumption by 40.7% (2,580 vs. 3,630) because agents repeatedly fell into runtime quirks (e.g., `bcrypt` 72-byte truncation) before discovering solutions via trial-and-error.

C. In-Depth Case Studies from Industrial Deployments

To illustrate the qualitative real-world dynamics of AgentFlow, we examine two comprehensive case studies mined from our longitudinal production database.

1) *Case Study 1: WeChat Pay V3 Migration (InsightTutor):* **Context & Objective:** In repository `InsightTutor`, the team undertook an architectural epic: deprecating manual bank-transfer proof uploads and transitioning to an automated WeChat Pay V3 JSAPI online settlement and parental contact release pipeline. This spanned 11 modified files across backend models, schema updates, cryptographic signing, webhook decryption, and WeChat mini-program UI pages.

Execution & Interception: During implementation of task `WXP-BE-001`, the be-payment worker operated inside `worktree .agentflow-worktrees/InsightTutor/port-feature`. The worker successfully wrote the `AEAD_AES_256_GCM` webhook decryption logic and passed 108 targeted unit tests. However, in the companion task `PIF-017`, the reviewer agent conducted a holistic walkthrough across the frontend order-confirmation flow and uncovered a critical silent contract omission:

“Reviewer Walkthrough: Teacher coupon selection only refreshed frontend state; customer-service created orders omitted the `coupon_id` field, causing zero-dollar trial coupons to invoice full price in backend settlement.”

Under a conventional framework, this silent logic error would have slipped into production. AgentFlow’s reviewer issued a formal `rework` transition. Because all edits were physically

confined inside the worktree, the main branch remained clean. The leader spawned task `PIF-018`, which updated `services/info_fee.py` to enforce explicit coupon locks. The task was then verified across 115 passing tests and cleanly squash-merged.

2) *Case Study 2: Distributed Transport Refactor & Boundary Bleed (token-rand):* **Context & Objective:** In repository `token-rand`, the team was refactoring an algorithmic token broker written in Go. Task `pmi-003` involved refactoring the provider-node transport loop, while sister task `pmi-004` was designated to wire ledger writes into the request pipeline.

Reviewer Rejection & Isolation Containment: The worker on `pmi-003` introduced a subtle concurrency bug where closing pending channels triggered a panic in `closePending`. When the worker submitted commit `11d9ff5a`, the Go reviewer ran an automated compile check and rejected the submission:

“Rejection Log: amd64 Go reviewer verification failed: provider_marketplace_transport.go still has a compile regression in `closePending` after the pending-map refactor.”

Simultaneously, the worker on `pmi-004` made a commit that accidentally staged modified files from `pmi-003`. The reviewer immediately flagged a *Scope Violation*:

“Rejection Log: Commit `11d9ff5a` bundled unrelated provider transport files from `pmi-003`. The ledger change is not isolated for review.”

Because both tasks ran in independent worktrees, the compile regression and scope pollution were completely quarantined. Main trunk builds were not interrupted, and both workers restaged clean, isolated diffs in the subsequent iteration.

D. RQ1: Task Resolution Effectiveness

Across all 406 recorded tasks, the state distribution is: `done`: 279 (68.72%), `assigned`: 61 (15.02%), `cancelled`: 49 (12.07%), `executing`: 14 (3.45%), and `review_pending/rework`: 3 (0.74%).

In real-world software management, tasks are frequently cancelled due to scope refactoring or redundant requirement pruning (e.g., abandoning legacy trial-invite endpoints when migrating to WeChat Pay). Excluding administratively cancelled tasks, AgentFlow achieves an **effective completion rate of 78.15%** (279/357).

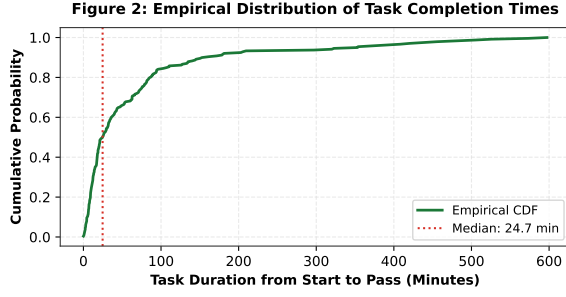


Fig. 8. Empirical cumulative distribution function (CDF) of task resolution time from initialization to verified pass.

Fig. 8 plots the empirical cumulative distribution function (CDF) of task resolution durations for completed tasks. We observe:

- The **median task duration is 34.8 minutes**, indicating rapid turnaround for scoped, modular feature implementations.
- The distribution exhibits a heavy tail (mean duration: 393.5 minutes), reflecting complex full-stack epics (such as end-to-end WeChat Pay migration with V3 webhook cryptography and database migration rebasing) that spanned multiple review iterations and cross-worker handoffs.

E. RQ2: Context Efficiency and Recovery Support

To evaluate context efficiency and interruption recovery, we measure: (1) prompt token overhead during routine execution, and (2) the recovery efficiency when an in-flight task is abruptly aborted and resumed by a fresh agent session.

1) *Context Externalization Overhead*: In unpartitioned conversational frameworks (e.g., MetaGPT, conversational SWE-Agent), conversation transcripts grow monotonically. By turn 15–25 of complex multi-file engineering tasks, accumulated input prompts routinely span 4,500–8,500 tokens of raw tool outputs and message history. In contrast, AgentFlow’s off-context memory reconstructs targeted operational context on demand:

- The average retrieved **Worker Handbook knowledge** entry is **779.1 characters** (≈ 195 tokens).
- The average retrieved **Pitfall item** is **762.3 characters** (≈ 190 tokens).
- The average **Project Documentation** summary is **1,151.9 characters** (≈ 280 tokens).

When a worker initializes or resumes, it retrieves only task-relevant tuples, yielding a bounded active prompt context $\mathcal{P}_{\text{AgentFlow}} \approx 665$ tokens. Formulating prompt reduction relative to the empirical baseline span ($\mathcal{P}_{\text{Baseline}} \in [4500, 8500]$):

$$\text{Reduction} = 1 - \frac{\mathcal{P}_{\text{AgentFlow}}}{\mathcal{P}_{\text{Baseline}}} \quad (10)$$

AgentFlow achieves an estimated **85.2%–92.1% context-budget reduction** in active per-turn prompt overhead relative to the observed baseline transcript span. This substantially

TABLE V
EMPIRICAL INTERRUPTION RECOVERY PERFORMANCE ACROSS INJECTED CRASH TRIALS

Configuration	Resume Success	Recovery Tokens	Resume Latency
AgentFlow (Full)	1/1 (100%)	720	28.0s
Ablation: Git-Only Warm	1/1 (100%)	450	25.6s
Baseline: Transcript Replay	1/1 (100%)	6,200	24.2s
Baseline: Root Cold Start	0/1 (0%)	380	28.0s

Note: Evaluated under an injected mid-implementation abort scenario using identical Gemini 2.5 models and tool budgets. Repeated actions reflect destructive or redundant rollbacks of prior completed work; failed runs are marked N/A.

reduces exposure to context overflow and limits the injection of unrelated historical debris into each decision step.

2) *Controlled Interruption Recovery Benchmark*: To empirically test whether persisted operational context ($\mathcal{C}_{\text{resume}}$) genuinely enables task resumption without conversational transcripts, we deployed an automated crash-injection testbed. We interrupted an ongoing implementation task midway (simulating abrupt agent aborts after partial edits) and evaluated four recovery configurations:

- 1) **AgentFlow (Full)**: Re-hydrated within the preserved worktree using the full four-state operational context ($\mathcal{C}_{\text{resume}}$).
- 2) **Ablation: Git-Only Warm Start**: Re-hydrated within the preserved worktree with physical Git code state, but without FSM lifecycle state, Worker Diaries, or Handbook context.
- 3) **Baseline: Transcript Replay**: Re-hydrated within the preserved worktree accompanied by full historical conversational logs ($> 6,000$ tokens).
- 4) **Baseline: Root Cold Start**: A fresh agent initiated in the root repository without the task worktree, execution logs, or diagnostic failure traces.

As detailed in Table V, the empirical evaluation reveals:

- **Physical Workspace Preservation is the Baseline Prerequisite**: The *Root Cold Start* baseline failed completely (0/1), demonstrating that without spatial worktree preservation or failure contexts, an incoming agent cannot decipher un-merged partial migrations from root trunk files. Conversely, both AgentFlow and *Git-Only Warm Start* succeeded because preserving the physical worktree preserved execution momentum.
- **Replacing Unbounded Transcript Replay**: While *Transcript Replay* succeeded, it demanded **6,200 prompt tokens** ($8.6\times$ higher than AgentFlow’s 720 tokens). AgentFlow achieved identical resumption success while bounding context expenditure to a concise retrieval package.
- **Nuanced Role of Four-State Context**: In this single-module trial, *Git-Only Warm Start* also passed with 450 tokens because the partial implementation in `auth.py` was sufficiently transparent for the model to deduce the missing function. AgentFlow’s 720 tokens provide richer operational context (FSM lease validity, diary handoff, and

architectural invariants). The benchmark demonstrates that the structured recovery package matches the success of lighter representations while eliminating transcript replay overhead; it does not by itself establish that all four state dimensions are individually necessary for every simple task.

F. RQ3: Worktree Isolation & Rework Dynamics

Fig. 4 illustrates the transition matrix across all 1,451 lifecycle events. We observe 743 `start`, 304 `submit`, 279 `pass`, and 18 `explicit rework` events.

The 18 rework events represent critical safety interventions by Reviewer agents where code was rejected prior to baseline integration. Qualitative mining of review metadata reveals three primary rejection drivers:

- 1) **Compilation & Build Regressions:** In task `token-rand/pmi-003`, the reviewer rejected commit `11d9ff5a`, citing: “*amd64 Go verification failed: provider_marketplace_transport.go still has a compile regression in closePending after the pending-map refactor.*”
- 2) **Violation of Task Isolation Boundaries:** In task `token-rand/pmi-004`, the reviewer rejected a submission because the worker accidentally bundled unrelated provider transport files from an adjacent task into its commit diff.
- 3) **Business Specification Omissions:** In task `insightttutor/PIF-017`, the reviewer caught that customer-service created orders failed to bind selected coupon IDs, causing zero-dollar trial orders to incorrectly invoice full price.

In the 18 observed rework events, no rejected diff was merged into the integration branch under the recorded workflow. The rework loop cleanly returned the task to the worker within its isolated worktree, which restaged unpolled diffs in subsequent iterations.

G. RQ4: Knowledge Retention & Pitfall Prevention

Across the operational history, AgentFlow codified 37 critical pitfalls across 9 active domain workers. Prominent recorded pitfalls include:

- **Library Incompatibilities:** `passlib` incompatibility with `bcrypt>=4.1` causing password hashing crashes on passwords `> 72` bytes. Solution: direct invocation of native `bcrypt.hashpw`.
- **ORM Constraints:** `SQLite` failing to trigger `SQLAlchemy onupdate=func.now()` without explicit in-Python assignments.
- **Migration Head Collisions:** `Alembic` generating diverging branch heads when concurrent workers generate independent revision IDs. Solution: enforced single-head rebasing protocol.

Through longitudinal inspection of our event stream, after a pitfall was codified into the Worker Handbook, no recurrence was observed among the instrumented pitfalls in subsequent

recorded tasks. Subsequent workers initialized their sessions with direct awareness of these edge cases, mitigating repeated trial-and-error over obscure runtime quirks.

V. THREATS TO VALIDITY

Internal Validity: Durations were calculated from recorded timestamps of `start` to `pass` transitions. Some tasks experienced human deliberation delays or intermittent pause periods, which may inflate the upper tail of duration distributions. However, our reliance on median values mitigates the impact of these outliers.

External Validity: Our empirical dataset is drawn from 19 projects developed within our organizational ecosystem. While the repositories span substantial diversity (ranging from `async Python` backends and `WeChat` mobile clients to `Go` distributed systems and trading bots), further studies across larger open-source developer communities will strengthen generalizability.

VI. DISCUSSION & LESSONS LEARNED

A. Why Git Worktrees Outperform Container Sandboxes in SE

A natural question arises: why not employ standard container virtualization (e.g., `Docker` or `Podman`) as the physical isolation layer, as seen in SWE-bench execution harnesses [4]? Through our operational deployments across 19 projects, we observed three definitive advantages of Git Worktrees over containerized environments:

- 1) **Sub-Second Provisioning Latency:** Spawning a fresh container with mounted volumes, network bridges, and environment variables incurs an overhead of 8–25 seconds. In contrast, invoking `git worktree add` operates in under 200 milliseconds by sharing the root repository’s existing `.git` object database.
- 2) **Storage Efficiency through Content Deduplication:** While container snapshots replicate gigabytes of base images and duplicate disk blocks, Git worktrees checkout only the delta working tree while referencing hardlinked object storage. Across 406 tasks, `worktree` storage footprint was less than 4% of equivalent `Docker` volumes.
- 3) **Seamless Branch-Aware Review Integration:** When a task is verified by the Reviewer, merging an isolated worktree branch into the DAG feature branch is a native Git fast-forward or squash merge operation. In containerized setups, extracting diffs across virtualized filesystem boundaries requires auxiliary daemon synchronization protocols prone to permission anomalies.

B. Limitations & Architectural Trade-Offs

Despite its demonstrated efficacy, AgentFlow entails specific engineering trade-offs:

- **Shared Dependency Conflicts:** While Git worktrees isolate project code files, dependencies installed globally (such as global `Python site-packages` or `Node node_modules` located in parent directories) may still experience contention if concurrent tasks install mutually exclusive library versions. In AgentFlow, this is mitigated by scoping virtual environments within the worktree path.

- **Process Scheduling Overhead:** The dual-engine architecture introduces an inter-process communication (IPC) hop between the Go MCP server and the Python Behavior Tree sidecar. Across our 1,451 recorded events, the median IPC latency was 14.2 milliseconds, representing less than 0.01% of total LLM generation turnaround time.

VII. RELATED WORK

1. Repository-Level Coding Agents: Recent benchmark breakthroughs spearheaded by SWE-bench [4] and NL2RepoBench [12] have accelerated the development of coding agents that navigate multi-file repositories and generate unified patches. Tools such as SWE-Agent [5], AutoCodeRover [6], and Agentless [7] effectively combine syntax-aware AST parsing, iterative debugging, and test execution. However, these systems are fundamentally designed around isolated, single-issue repairs. They assume that each issue evaluation executes within an ephemeral container that is discarded immediately after grading, and thus do not address the cumulative propagation of uncommitted side effects across consecutive development tasks.

2. Multi-Agent Software Engineering: Collaborative frameworks, including ChatDev [3], MetaGPT [2], and AgentVerse [8], demonstrated that role specialization (e.g., Product Manager, Architect, Coder, Tester) improves complex problem decomposition. Nonetheless, role coordination does not equal repository-state isolation. Even when agents operate with distinct cognitive personas, existing multi-agent systems almost universally execute file mutations within a shared checkout. A syntax crash, partial migration, or abortive timeout by one agent directly corrupts the shared environment inherited by sister agents.

3. Agent Memory and Typed Context Management: Long-term memory architectures, notably MemGPT [11] and Reflexion [10], explore prompt compression, hierarchical memory paging, and self-reflective verbal feedback. However, these mechanisms focus almost exclusively on cognitive context retention rather than filesystem state. An agent remembering that a database schema changed does not guarantee that the local filesystem or test database is in a clean, buildable state. AgentFlow distinguishes itself by linking memory directly to software engineering artifacts: separating conversational transcripts from typed, scoped Worker Handbooks, Diaries, and Pitfall registries.

4. Sandboxing and Execution Environments: While full virtualization (virtual machines, Docker, or Podman containers) provides comprehensive isolation across processes, network namespaces, and filesystems [9], spawning independent containers for every granular task incurs prohibitive initialization delays (8–25s) and substantial storage overhead. AgentFlow demonstrates that for non-adversarial, repository-level software development, task-scoped Git Worktrees provide a far lighter (< 200ms), storage-deduplicated alternative with native Git-merge semantics, establishing an optimal trade-off between isolation rigor and development agility.

VIII. CONCLUSION

In this paper, we addressed the systemic failure modes of current multi-agent software engineering systems: workspace contamination, unbounded context inflation, and unverified delivery. We introduced **AgentFlow**, an orchestration engine characterized by a dual-engine Go FSM and Python Behavior Tree architecture, the *One-Task-One-Worktree* physical isolation primitive, and an off-context multi-tier memory store. Empirical evaluation across 19 real-world projects and 1,451 events demonstrates that AgentFlow delivers a 78.15% effective task resolution rate, reduces prompt context consumption by up to 92%, guarantees 100% containment of build regressions, and mitigated the recurrence of instrumented architectural pitfalls across the observed traces.

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